

STBL ESS / Redemptions Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Stalin

Alix40

June 17, 2026

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **4**

7 Risk Classification **4**

8 Protocol Summary **4**

8.1 ESS 4

8.2 EVM Redemptions 5

9 Attestation **5**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for STBL on STBL ESS / Redemptions. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated June 17, 2026, was delivered to STBL, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from STBL or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 8 days, the Cyfrin team conducted an audit on the STBL ESS / Redemptions code provided by STBL. The findings consist of 2 Critical, 8 High, 8 Medium & 9 Low severity issues with the remainder being informational.

Summary

Project Name	STBL ESS / Redemptions
Repository	stbl-contracts-evm-ess-common
Commit	520f6ae76831...
Fix Commit	6a3df6e5967e...
Repository 2	stbl-contracts-evm-redemptions
Commit	f749cd799296...
Fix Commit	37d5ee7b9fea...
Audit Timeline	May 18th - May 27th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	2	2	0
High Risk	8	8	0
Medium Risk	8	8	0
Low Risk	9	8	1
Informational	9	4	5
Gas Optimizations	0	0	0
Total	36	30	6

6 Audit Scope

ESS: the audit scope was limited to the following files of stbl-contracts-evm-ess-common:

```
stbl-contracts-evm-ess-common/contracts/nft-vault/STBL_ESS_NFT_Vault1.sol
stbl-contracts-evm-ess-common/contracts/wrapper/STBL_ESS_Wrapper1.sol
stbl-contracts-evm-ess-common/contracts/test/assets/STBL_XLayer_Asset_Issuer.sol
stbl-contracts-evm-ess-common/contracts/test/assets/STBL_XLayer_Asset_Oracle.sol
stbl-contracts-evm-ess-common/contracts/test/assets/STBL_XLayer_Asset_Vault.sol
stbl-contracts-evm-ess-common/contracts/test/assets/STBL_XLayer_Asset_YieldDistributor.sol
stbl-contracts-evm-ess-common/contracts/test/splitter/STBL_YLD_SplitMerge.sol
stbl-contracts-evm-ess-common/contracts/test/STBL_XLayer_NFT_Vault.sol
stbl-contracts-evm-ess-common/contracts/test/STBL_XLayer_Wrapper.sol
stbl-contracts-evm-ess-common/contracts/test/STBL_XLLayer_Token.sol
```

EVM-Redemptions: the audit scope was limited to the following files of stbl-contracts-evm-redemptions:

```
stbl-contracts-evm-redemptions/contracts/redemption/STBL_Redemption_Core.sol
stbl-contracts-evm-redemptions/contracts/redemption/STBL_Redemption.sol
stbl-contracts-evm-redemptions/contracts/splitter/STBL_YLD_SplitMerge.sol
stbl-contracts-evm-redemptions/contracts/Lock.sol
stbl-contracts-evm-redemptions/contracts/Util.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

8.1 ESS

The ESS module is a multi-asset deposit facility for the STBL stablecoin ecosystem. It allows users to deposit multiple real-world-asset (RWA)-backed tokens simultaneously in predefined ratios and receive a single composite `ESS` ERC-20 stablecoin in return. Each deployment configures a fixed basket (for example 30% of asset A and 70% of asset B) and the wrapper enforces those ratios on every deposit.

Each individual asset deposit is processed through its own `iSTBL_Issuer` contract, which accepts the asset token, charges fees and mints a `YLD NFT` recording the position's value. A single ESS deposit therefore produces one YLD NFT per configured asset; this set of NFTs is grouped into a lot held by the `STBL_ESS_NFT_Vault` for the life of the position.

Withdrawal reverses the process: the user returns their ESS tokens to the wrapper, which routes through the vault to burn each NFT via its issuer, releasing the underlying asset tokens. Between deposit and withdrawal, the YLD NFTs accumulate yield through per-asset YieldDistributor contracts.

Users can also combine two of their lots into one, which merges each pair of NFTs into a single NFT carrying the combined value.

The concrete deployment (`STBL_XLayer_Wrapper`, `STBL_XLayer_NFT_Vault`, `STBL_XLayer_Token`) targets the XLayer network and adds ERC-2771 meta-transaction support.

8.2 EVM Redemptions

STBL is a multi-asset-backed stablecoin system where users deposit real-world asset (RWA) tokens into per-asset issuers, which mint YLD NFTs: ERC-721 tokens encoding each depositor's position as structured metadata (principal value, fee snapshot, yield-staking state). Multiple YLD NFTs can be bundled into a basket via the ESS module to mint USST, the protocol's stablecoin.

The Redemptions module is a single-asset liquidity pool that provides the exit path: USST holders can exchange their stablecoins for underlying RWA tokens without originating a full ESS position, while liquidity providers earn passive yield on the positions they contribute to the pool. It connects two participant groups:

1. Liquidity providers (LPs) deposit YLD NFTs into the pool to earn yield from the underlying assets and can withdraw their proportional share.
2. Redeemers are USST stablecoin holders who burn USST against the pool's NFT, receiving the underlying RWA tokens in return.

The pool is implemented as a single aggregated position: LP deposits are merged into one pool YLD NFT, each LP holds a pro-rata share of its value, and yield pulled from the underlying `iSTBL_YieldDistributor` is shared among LPs in proportion to their deposits.

The four primary operations are:

- **Deposit**: an LP adds a YLD NFT to the pool position.
- **Withdraw**: an LP takes back a YLD NFT for part or all of their share.
- **Claim yield**: yield is collected from the asset's yield distributor and LPs claim their earned share.
- **Redeem**: a USST holder burns USST against the pool position and receives the underlying RWA tokens.

`STBL_YLD_SplitMerge` handles all NFT split and merge operations.

9 Attestation

Cyfrin attests that it performed a security review of the STBL ESS / Redemptions code provided by STBL.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-06-17-cyfrin-stbl-ess-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to STBL has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e4cced774b51c30f84c9c31e9f81e2f3898f046e0fff7c962357a27e0f308a9b
F=2026-06-17-cyfrin-stbl-ess-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.